# Generic3 model

## Parametrization

The generic3 model implements the following precision matrix

$$\mathbf{Q} = \tau_{\text{common}} \sum_{i=1}^{m} \tau_i \boldsymbol{R}_i, \qquad 1 < m \leq 10, \tag{1}$$

where each $\boldsymbol{R}_i \geq 0$, and $\tau_i$ is the specific precision parameter and $\tau_{\text{common}}$ is a shared one.

## Hyperparameters

The hyperparameters are defined as

$$\theta_i = \log(\tau_i), \qquad i = 1, 10$$

and

$$\theta_{11} = \log(\tau_{\text{common}})$$

and priors are assigned to $(\theta_1, \theta_2, \ldots)$.

## Specification

The generic3 model is specified inside the `f()` function as

```
f(<whatever>, model="generic3", Cmatrix = <list.of.Cmat>, hyper = <hyper>)
```

where `<list.of.Cmat>` a list of length $m$ (maximum 10) of $\boldsymbol{R}_i$-matrices. By default, $\theta_j$ for $j = m+1, \ldots, 11$ is set to fixed (this includes $\tau_{\text{common}}$).

### Hyperparameter spesification and default values

**doc** `A generic model (type 3)`

**hyper**

- **theta1**
  - **hyperid** `21001`
  - **name** `log precision1`
  - **short.name** `prec1`
  - **initial** `4`
  - **fixed** `FALSE`
  - **prior** `loggamma`
  - **param** `1 5e-05`
  - **to.theta** `function(x) log(x)`
  - **from.theta** `function(x) exp(x)`
- **theta2**
  - **hyperid** `21002`
  - **name** `log precision2`
  - **short.name** `prec2`
  - **initial** `4`
  - **fixed** `FALSE`

**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta3**

**hyperid** 21003
**name** log precision3
**short.name** prec3
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta4**

**hyperid** 21004
**name** log precision4
**short.name** prec4
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta5**

**hyperid** 21005
**name** log precision5
**short.name** prec5
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta6**

**hyperid** 21006
**name** log precision6
**short.name** prec6
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta7**

**hyperid** 21007
**name** log precision7
**short.name** prec7
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta8**

**hyperid** 21008
**name** log precision8
**short.name** prec8
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta9**

**hyperid** 21009
**name** log precision9
**short.name** prec9
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta10**

**hyperid** 21010
**name** log precision10
**short.name** prec10
**initial** 4
**fixed** FALSE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**theta11**

**hyperid** 21011
**name** log precision common
**short.name** prec.common

**initial** 0
**fixed** TRUE
**prior** loggamma
**param** 1 5e-05
**to.theta** function(x) log(x)
**from.theta** function(x) exp(x)

**constr** FALSE

**nrow.ncol** FALSE

**augmented** FALSE

**aug.factor** 1

**aug.constr**

**n.div.by**

**n.required** TRUE

**set.default.values** TRUE

**pdf** generic3

## Example

```
make.sm = function(n, prob = 0.1)
{
    A = matrix(runif(n*n), n, n)
    Patt = matrix(runif(n*n), n, n)
    Patt[Patt > prob] = 0
    A = A * Patt
    diag(A) = runif(n)
    A = A %*% t(A)
    A = A / max(diag(A))
    return (inla.as.sparse(A))
}

nsim = 100
n = 5
m = 3
Cmat = list()
Q = inla.as.sparse(matrix(0, n, n))
for(i in 1:m) {
    Cmat[[i]] = make.sm(n)
    Q = Q + i*Cmat[[i]]
}

yy = inla.qsample(nsim, Q)
y = c(yy)
idx = rep(1:n, nsim)
r = rep(1:nsim, each = n)

r = inla(y ~ -1 +
    f(idx, model="generic3",
      Cmatrix = Cmat, replicate = r,
```

```
      hyper = list(
          prec1 = list(initial = log(1)),
          prec2 = list(initial = log(2)),
          prec3 = list(initial = log(3)))),
    data = list(y=y, Cmat=Cmat, r=r),
    verbose=TRUE,
    control.family = list(
        hyper = list(
            prec = list(
                initial = 10,
                fixed = TRUE))))
summary(r)
```

## Notes